Anesthesia for Hepatic Disease V2 – MCQ

Source: Short Textbook of Anesthesia – Ajay Yadav

Anesthesia for Hepatic Disease

50 Questions • 30 Minutes

Instructions:
• Each question has 4 options (A, B, C, D).
• The correct answer is shown on the right side of each question.
• Detailed explanations with textbook page references are at the end.

Questions

Q1. What is the gold standard scoring system to assess the severity of hepatic disease and risk assessment? **A.** MELD score **B.** Child-Pugh classification **C.** APACHE score **D.** Glasgow Coma Scale	**Ans: B**
Q2. The aim of preoperative evaluation in hepatic patients is to assess the severity of disease AND: **A.** Only the liver enzymes **B.** The damage caused to other organs by hepatic dysfunction **C.** Only the coagulation profile **D.** Only the nutritional status	**Ans: B**
Q3. In Child-Pugh classification, serum bilirubin value for Group C is: **A.** < 2 mg% **B.** 2–3 mg% **C.** > 3 mg% **D.** > 5 mg%	**Ans: C**
Q4. In Child-Pugh classification, serum albumin for Group A is: **A.** < 3 gm% **B.** 3–3.5 gm% **C.** > 3.5 gm% **D.** > 4.5 gm%	**Ans: C**
Q5. Child-Pugh Class C has a mortality of: **A.** 2–5% **B.** 10% **C.** > 50% **D.** 25%	**Ans: C**
Q6. Hepatic patients should always be screened for: **A.** HIV only **B.** Hepatitis B and C **C.** Hepatitis A only **D.** Tuberculosis	**Ans: B**
Q7. When should elective surgery be deferred in acute liver disease? **A.** For a fixed period of 6 weeks **B.** Till the liver functions are stabilized **C.** For 3 months always **D.** Only if bilirubin > 10	**Ans: B**

Q8. Patients with moderate to severe chronic hepatic disease should have restoration of coagulation abnormalities by: **A.** Heparin infusion **B.** Vitamin K, cryoprecipitate or fresh frozen plasma **C.** Warfarin **D.** Aspirin discontinuation only	**Ans: B**
Q9. Premedication with benzodiazepines in hepatic patients: **A.** Is the preferred premedicant **B.** Should not be given as hepatic patients are oversensitive to sedative effects **C.** Can be given in full dose **D.** Is mandatory for anxiolysis	**Ans: B**
Q10. Why is central venous pressure (CVP) monitoring highly recommended in hepatic patients? **A.** To measure cardiac output **B.** Due to fluid loss to 3rd space and intravascular volume depletion **C.** To administer drugs centrally **D.** To monitor for air embolism	**Ans: B**
Q11. Due to high volume of distribution in hepatic patients, the initial dose of intravenous agents is: **A.** Reduced **B.** More (higher than normal) **C.** Same as normal patients **D.** Half the normal dose	**Ans: B**
Q12. In hepatic patients, repeated/maintenance doses of IV agents may be prolonged because of: **A.** Increased renal excretion **B.** Decreased metabolism **C.** Increased protein binding **D.** Enhanced biliary excretion	**Ans: B**
Q13. The induction agent of choice for hepatic patients is Propofol because: **A.** It has the longest duration **B.** It has short half-life and extrahepatic metabolism **C.** It is completely metabolized by the liver **D.** It increases hepatic blood flow	**Ans: B**
Q14. The inhalational agent of choice for hepatic patients is: **A.** Halothane **B.** Isoflurane **C.** Sevoflurane **D.** Nitrous oxide	**Ans: C**

Q15. The most important goal during maintenance of anesthesia in hepatic patients is: **A.** Maintaining adequate depth of anesthesia **B.** Maximum perseverance of hepatic blood flow **C.** Keeping heart rate stable **D.** Avoiding awareness	**Ans: B**

Q16. Halothane must not be used in hepatic patients for TWO reasons: **A.** Seizures and bronchospasm **B.** May cause hepatitis AND causes significant decrease in hepatic and portal blood flow **C.** Renal toxicity and cardiac depression **D.** Malignant hyperthermia and hyperkalemia	**Ans: B**

Q17. Nitrous oxide should be avoided in hepatic patients for TWO reasons: **A.** It causes seizures and hepatitis **B.** It can increase pulmonary artery pressure (portopulmonary hypertension) AND requires higher O2 concentration (hepatopulmonary syndrome) **C.** It causes renal failure and coagulopathy **D.** It causes hypoglycemia and encephalopathy	**Ans: B**

Q18. Why are desflurane and isoflurane considered good alternatives to sevoflurane in hepatic patients? **A.** They are cheaper **B.** They preserve hepatic blood flow and produce minimal trifluoroacetic acid (hepatotoxicity is just theoretical) **C.** They have the strongest analgesic properties **D.** They do not require a vaporizer	**Ans: B**

Q19. Succinylcholine can produce prolonged block in hepatic patients because of: **A.** Increased sensitivity of receptors **B.** Decreased pseudocholinesterase **C.** Increased volume of distribution **D.** Electrolyte imbalance	**Ans: B**

Q20. The muscle relaxants of choice for hepatic patients are: **A.** Vecuronium and Pancuronium **B.** Atracurium and Cis-atracurium **C.** Rocuronium and Vecuronium **D.** Succinylcholine and Mivacurium	**Ans: B**

Q21. Atracurium and Cis-atracurium are preferred in hepatic patients because they are metabolized by: **A.** Hepatic cytochrome P450 **B.** Hoffman degradation **C.** Renal excretion **D.** Biliary excretion	**Ans: B**

Q22. Remifentanil is the opioid of choice in hepatic patients because:

A. It has the longest duration
B. It is metabolized by plasma esterases
C. It is excreted unchanged by kidneys
D. It has no respiratory depression

Ans: B

Q23. In hepatic patients, decreased metabolism of citrate (from blood transfusion) means these patients should receive:

A. Magnesium supplementation
B. Calcium after blood transfusion
C. Potassium replacement
D. Sodium bicarbonate

Ans: B

Q24. Which of the following is NOT acceptable intraoperatively in hepatic patients?

A. Normothermia
B. Hypotension
C. Adequate urine output
D. Normal blood glucose

Ans: B

Q25. Albumin is the preferred colloid for hepatic patients because hypotension:

A. Causes bronchospasm
B. Can severely impair hepatic blood flow
C. Increases portal pressure
D. Causes seizures

Ans: B

Q26. Alkalosis is dangerous in hepatic patients because it:

A. Causes hypokalemia
B. Increases the conversion of ammonium to ammonia which crosses blood brain barrier to precipitate hepatic encephalopathy
C. Causes hypocalcemia only
D. Decreases cardiac output

Ans: B

Q27. Hepatic disease patients are prone for hypoglycemia because:

A. They have excessive insulin production
B. They have impaired gluconeogenesis and exhausted glucose stores
C. They have increased glucose utilization
D. They have renal glucose wasting

Ans: B

Q28. Sepsis in hepatic patients is dangerous because it can precipitate:

A. Bronchospasm
B. Hepatic encephalopathy
C. Malignant hyperthermia
D. Anaphylaxis

Ans: B

Q29. Decreased urinary output in hepatic patients can precipitate: **A.** Pulmonary edema **B.** Hepatorenal syndrome **C.** Cardiac failure **D.** Cerebral edema	**Ans: B**
Q30. Postoperatively in hepatic patients, which functions should be closely monitored? **A.** Only cardiac function **B.** Renal functions and urine output **C.** Only respiratory function **D.** Only neurological function	**Ans: B**
Q31. Cirrhotic patients have intrapulmonary shunt, therefore postoperatively they are more prone for: **A.** Hyperoxia **B.** Hypoxia **C.** Hypercarbia **D.** Metabolic alkalosis	**Ans: B**
Q32. Due to existing coagulopathy and intravascular volume depletion, which anesthetic technique should be avoided in hepatic patients? **A.** General anesthesia **B.** Central neuraxial blocks **C.** Peripheral nerve blocks **D.** Local infiltration	**Ans: B**
Q33. When can central neuraxial block be considered in hepatic patients? **A.** Never **B.** If there is no associated coagulopathy and hypotension can be prevented **C.** Only in Child-Pugh A **D.** Only with platelet transfusion cover	**Ans: B**
Q34. Patients with biliary obstruction exhibit bradycardia because of: **A.** Vagal stimulation from pain **B.** Direct effect of bile salts on Sinoatrial (SA) node **C.** Hypothermia **D.** Beta-blocker effect of bilirubin	**Ans: B**
Q35. Just after the release of biliary obstruction, which drug should be given to wash out the bile salts? **A.** Furosemide **B.** Mannitol **C.** Dexamethasone **D.** Normal saline bolus	**Ans: B**

Q36. Vecuronium should not be used in patients with biliary obstruction because: **A.** It causes histamine release **B.** It has 30–40% excretion through bile **C.** It causes bradycardia **D.** It is metabolized to toxic metabolites	**Ans: B**

Q37. Regarding opioids and sphincter of Oddi spasm, the textbook states that the incidence of opioid-induced spasm is: **A.** More than 20% **B.** Around 10% **C.** Less than 3% **D.** More than 50%	**Ans: C**

Q38. The strict guideline of not using opioids for biliary colic has been: **A.** Reinforced by recent studies **B.** Questioned in current day practice **C.** Made into a law **D.** Unchanged for decades	**Ans: B**

Q39. In Child-Pugh classification, prothrombin time prolongation from control for Group B is: **A.** 1–4 seconds **B.** 4–6 seconds **C.** > 6 seconds **D.** No prolongation	**Ans: B**

Q40. In Child-Pugh classification, ascites in Group A is: **A.** Moderate **B.** Marked **C.** None **D.** Mild	**Ans: C**

Q41. Child-Pugh Group B has a surgical risk of: **A.** Minimal **B.** Moderate **C.** Marked **D.** None	**Ans: B**

Q42. Why should sevoflurane be preferred over halothane — what metabolite does sevoflurane NOT produce? **A.** Fluoride ions **B.** Trifluoroacetic acid **C.** Compound A **D.** Carbon monoxide	**Ans: B**

Q43. Encephalopathy in hepatic patients should be corrected preoperatively by:

- **A.** Phenytoin and carbamazepine
- **B.** Lactulose and neomycin
- **C.** Mannitol and furosemide
- **D.** Dexamethasone and hypertonic saline

Ans: B

Q44. If ascites is significant enough to impair respiratory functions, what should be done preoperatively?

- **A.** Diuretics only
- **B.** Tapping of ascites
- **C.** TIPS procedure
- **D.** Paracentesis is contraindicated

Ans: B

Q45. Child-Pugh classification includes all of the following parameters EXCEPT:

- **A.** Encephalopathy
- **B.** Nutrition
- **C.** Creatinine
- **D.** Ascites

Ans: C

Q46. Decreased albumin in hepatic patients may increase the unbound fraction of which aspect of drug pharmacology?

- **A.** Drug absorption
- **B.** Drug binding — increasing the free fraction of the drug
- **C.** Drug excretion
- **D.** Drug distribution volume only

Ans: B

Q47. Universal precautions intraoperatively in hepatic patients must be taken if the patient is positive for:

- **A.** H. pylori
- **B.** Hepatitis B or C
- **C.** CMV
- **D.** EBV

Ans: B

Q48. Preoperative evaluation of hepatic patients should rule out all of the following EXCEPT:

- **A.** Coagulopathy and renal damage (hepatorenal syndrome)
- **B.** Pulmonary dysfunction (hepatopulmonary syndrome)
- **C.** Rheumatoid arthritis
- **D.** Portopulmonary hypertension and encephalopathy

Ans: C

Q49. In the management of biliary obstruction patients, what is stated about the general principles of hepatic disease management?

- **A.** Completely different guidelines apply
- **B.** Whatever discussed for hepatic disease is applicable, with additional points to be taken care
- **C.** Only regional anesthesia should be used
- **D.** No special considerations needed

Ans: B

Q50. **Alcoholism should always be ruled out in hepatic patients because of associated:** **A.** Only liver damage **B.** Comorbidities including cardiomyopathy, pancreatitis, and nutritional deficiencies **C.** Only psychiatric issues **D.** Only renal damage	**Ans: B**

Answer Key & Explanations

Q1. Answer: B — Child-Pugh classification

Correct Answer: B. Child-Pugh classification — Child-Pugh classification is the gold standard scoring system to assess the severity of hepatic disease and hence the risk assessment (Table 21.1).

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Preoperative Evaluation

Q2. Answer: B — The damage caused to other organs by hepatic dysfunction

Correct Answer: B. The damage caused to other organs by hepatic dysfunction — The aim is not only to assess severity but also to assess the damage caused to other organs by hepatic dysfunction: coagulopathy, renal damage (hepatorenal syndrome), pulmonary dysfunction (hepatopulmonary syndrome), portopulmonary hypertension, encephalopathy, ascites, cardiac functions, electrolyte and metabolic abnormalities.

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Preoperative Evaluation

Q3. Answer: C — > 3 mg%

Correct Answer: C. > 3 mg% — In Child-Pugh classification: Group A: bilirubin < 2, Group B: 2–3, Group C: > 3 mg%.

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Preoperative Evaluation (Table 21.1)

Q4. Answer: C — > 3.5 gm%

Correct Answer: C. > 3.5 gm% — Child-Pugh Group A: albumin > 3.5, Group B: 3–3.5, Group C: < 3 gm%.

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Preoperative Evaluation (Table 21.1)

Q5. Answer: C — > 50%

Correct Answer: C. > 50% — Per Table 21.1: Group A mortality 2–5%, Group B 10%, Group C > 50%.

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Preoperative Evaluation (Table 21.1)

Q6. Answer: B — Hepatitis B and C

Correct Answer: B. Hepatitis B and C — Always screen the hepatic patients for hepatitis B and C. Universal precautions must be taken intraoperatively if the patient is positive.

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Preoperative Evaluation

Q7. Answer: B — Till the liver functions are stabilized

Correct Answer: B. Till the liver functions are stabilized — Elective surgery should be deferred in acute liver disease till the liver functions are stabilized. No need to defer in asymptomatic patients with mildly elevated liver enzymes.

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Timing of Surgery

Q8. Answer: B — Vitamin K, cryoprecipitate or fresh frozen plasma

Correct Answer: B. Vitamin K, cryoprecipitate or fresh frozen plasma — Restoration of coagulation abnormalities (by vitamin K, cryoprecipitate or fresh frozen plasma) is essential before surgery.

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Timing of Surgery

Q9. Answer: B — Should not be given as hepatic patients are oversensitive to sedative effects

Correct Answer: B. Should not be given as hepatic patients are oversensitive to sedative effects — As hepatic patients may be oversensitive to the sedative effects of benzodiazepine, premedication with benzodiazepines should not be given.

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Premedication

Q10. Answer: B — Due to fluid loss to 3rd space and intravascular volume depletion

Correct Answer: B. Due to fluid loss to 3rd space and intravascular volume depletion — As there is fluid loss to 3rd space, the intravascular volume is depleted in these patients therefore, central venous pressure monitoring is highly recommended.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative – Monitoring

Q11. Answer: B — More (higher than normal)

Correct Answer: B. More (higher than normal) — Due to high volume of distribution the initial dose of intravenous agents is more while due to decreased metabolism the action of repeated/maintenance doses may be prolonged.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative – General Anesthesia

Q12. Answer: B — Decreased metabolism

Correct Answer: B. Decreased metabolism — While initial doses are higher (high Vd), the action of repeated/maintenance doses may be prolonged due to decreased metabolism. Also decreased albumin may increase the unbound fraction of drugs.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative – General Anesthesia

Q13. Answer: B — It has short half-life and extrahepatic metabolism

Correct Answer: B. It has short half-life and extrahepatic metabolism — Propofol, due to its short half-life and extrahepatic metabolism, is the induction agent of choice for hepatic patients.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative – Induction

Q14. Answer: C — Sevoflurane

Correct Answer: C. Sevoflurane — Sevoflurane is the inhalational agent of choice for two reasons: (1) Maximum perseverance of hepatic blood flow, (2) It does not produce Trifluoroacetic acid avoiding any possibility of hepatitis.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative – Maintenance

Q15. Answer: B — Maximum perseverance of hepatic blood flow

Correct Answer: B. Maximum perseverance of hepatic blood flow — Maintaining hepatic blood flow should be the most important goal in these patients.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative – Maintenance

Q16. Answer: B — May cause hepatitis AND causes significant decrease in hepatic and portal blood flow

Correct Answer: B. May cause hepatitis AND causes significant decrease in hepatic and portal blood flow — Halothane must not be used due to two reasons: firstly, it may cause hepatitis and secondly it causes significant decrease in hepatic as well as portal blood flow.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative – Maintenance

Q17. Answer: B — It can increase pulmonary artery pressure (portopulmonary hypertension) AND requires higher O2 concentration (hepatopulmonary syndrome)

Correct Answer: B. It can increase pulmonary artery pressure (portopulmonary hypertension) AND requires higher O2 concentration (hepatopulmonary syndrome) — Nitrous oxide should be avoided for two reasons: it can increase pulmonary artery pressure (portopulmonary hypertension) and it allows only lower oxygen concentration when higher concentration is needed due to significant right to left shunt (hepatopulmonary syndrome).

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative – Maintenance

Q18. Answer: B — They preserve hepatic blood flow and produce minimal trifluoroacetic acid (hepatotoxicity is just theoretical)

Correct Answer: B. They preserve hepatic blood flow and produce minimal trifluoroacetic acid (hepatotoxicity is just theoretical) — Desflurane (least metabolism) and isoflurane preserve hepatic blood flow. The amount of trifluoroacetic acid produced is too minimal that hepatotoxicity appears to be just theoretical.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative – Maintenance

Q19. Answer: B — Decreased pseudocholinesterase

Correct Answer: B. Decreased pseudocholinesterase — Succinylcholine can produce prolonged block due to decreased pseudocholinesterase synthesized by the liver.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative – Muscle Relaxants

Q20. Answer: B — Atracurium and Cis-atracurium

Correct Answer: B. Atracurium and Cis-atracurium — As atracurium and cis-atracurium are metabolized by Hoffman degradation, they are the muscle relaxants of choice for hepatic patients (organ-independent metabolism).

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative – Muscle Relaxants

Q21. Answer: B — Hoffman degradation

Correct Answer: B. Hoffman degradation — Hoffman degradation is organ-independent, making atracurium and cis-atracurium ideal for patients with hepatic dysfunction.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative – Muscle Relaxants

Q22. Answer: B — It is metabolized by plasma esterases

Correct Answer: B. It is metabolized by plasma esterases — Remifentanil is the opioid of choice because it is metabolized by plasma esterases, independent of hepatic function.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative – Muscle Relaxants

Q23. Answer: B — Calcium after blood transfusion

Correct Answer: B. Calcium after blood transfusion — As there is decreased metabolism of citrate (which can chelate calcium), these patients should receive calcium after blood transfusion.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative

Q24. Answer: B — Hypotension

Correct Answer: B. Hypotension — Hypotension can severely impair hepatic blood flow and is not acceptable intraoperatively. Other unacceptable conditions: alkalosis, hypoglycemia, sepsis, decreased urinary output.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Not Acceptable Intraoperatively

Q25. Answer: B — Can severely impair hepatic blood flow

Correct Answer: B. Can severely impair hepatic blood flow — Hypotension can severely impair hepatic blood flow. Albumin is the preferred colloid for a patient with compromised hepatic functions.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Not Acceptable Intraoperatively

Q26. Answer: B — Increases the conversion of ammonium to ammonia which crosses blood brain barrier to precipitate hepatic encephalopathy

Correct Answer: B. Increases the conversion of ammonium to ammonia which crosses blood brain barrier to precipitate hepatic encephalopathy — Alkalosis increases conversion of ammonium to ammonia which crosses blood brain barrier to precipitate hepatic encephalopathy.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Not Acceptable Intraoperatively

Q27. Answer: B — They have impaired gluconeogenesis and exhausted glucose stores

Correct Answer: B. They have impaired gluconeogenesis and exhausted glucose stores — Hepatic disease patients have impaired gluconeogenesis and exhausted glucose stores therefore, they are prone for hypoglycemia.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Not Acceptable Intraoperatively

Q28. Answer: B — Hepatic encephalopathy

Correct Answer: B. Hepatic encephalopathy — Sepsis can precipitate encephalopathy in hepatic patients. It is one of the conditions not acceptable intraoperatively.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Not Acceptable Intraoperatively

Q29. Answer: B — Hepatorenal syndrome

Correct Answer: B. Hepatorenal syndrome — Decreased urinary output can precipitate hepatorenal syndrome in hepatic patients and is not acceptable intraoperatively.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Not Acceptable Intraoperatively

Q30. Answer: B — Renal functions and urine output

Correct Answer: B. Renal functions and urine output — Postoperatively, renal functions and urine output should be monitored along with good antibiotic cover and oxygenation.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Postoperatively

Q31. Answer: B — Hypoxia

Correct Answer: B. Hypoxia — Cirrhotic patients have intrapulmonary shunt therefore, are more prone for hypoxia. Oxygenation should be closely monitored postoperatively.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Postoperatively – Oxygenation

Q32. Answer: B — Central neuraxial blocks

Correct Answer: B. Central neuraxial blocks — Due to the possibility of existing coagulopathy and intravascular volume depletion, central neuraxial blocks should be avoided in hepatic patients.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Choice of Anesthesia

Q33. Answer: B — If there is no associated coagulopathy and hypotension can be prevented

Correct Answer: B. If there is no associated coagulopathy and hypotension can be prevented — If there is no associated coagulopathy, central neuraxial block can be considered if hypotension can be prevented.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Choice of Anesthesia

Q34. Answer: B — Direct effect of bile salts on Sinoatrial (SA) node

Correct Answer: B. Direct effect of bile salts on Sinoatrial (SA) node — These patients exhibit bradycardia because of direct effect of bile salts on Sinoatrial (SA) node.

Topic: Anesthesia for Biliary Obstruction | Ref: Ch 21, Page 200, Topic: Anesthesia for Patients with Biliary Obstruction

Q35. Answer: B — Mannitol

Correct Answer: B. Mannitol — Just after the release of obstruction, mannitol should be given to wash out the bile salts.

Topic: Anesthesia for Biliary Obstruction | Ref: Ch 21, Page 200, Topic: Anesthesia for Patients with Biliary Obstruction

Q36. Answer: B — It has 30–40% excretion through bile

Correct Answer: B. It has 30–40% excretion through bile — Muscle relaxant vecuronium which has 30–40% excretion through bile should not be used in biliary obstruction patients.

Topic: Anesthesia for Biliary Obstruction | Ref: Ch 21, Page 200, Topic: Anesthesia for Patients with Biliary Obstruction

Q37. Answer: C — Less than 3%

Correct Answer: C. Less than 3% — The incidence of opioid-induced spasm of sphincter of Oddi is less than 3%, therefore strict guidelines of not using opioids for biliary colic has been questioned in current day practice.

Topic: Anesthesia for Biliary Obstruction | Ref: Ch 21, Page 200, Topic: Anesthesia for Patients with Biliary Obstruction

Q38. Answer: B — Questioned in current day practice

Correct Answer: B. Questioned in current day practice — As the incidence of opioid-induced sphincter of Oddi spasm is less than 3%, strict guidelines of not using opioids for biliary colic has been questioned in current day practice.

Topic: Anesthesia for Biliary Obstruction | Ref: Ch 21, Page 200, Topic: Anesthesia for Patients with Biliary Obstruction

Q39. Answer: B — 4–6 seconds

Correct Answer: B. 4–6 seconds — Per Table 21.1: Group A: PT prolongation 1–4 sec, Group B: 4–6 sec, Group C: > 6 seconds.

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Preoperative Evaluation (Table 21.1)

Q40. Answer: C — None

Correct Answer: C. None — Per Table 21.1: Group A: No ascites, Group B: Moderate, Group C: Marked.

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Preoperative Evaluation (Table 21.1)

Q41. Answer: B — Moderate

Correct Answer: B. Moderate — Per Table 21.1: Group A: Minimal surgical risk, Group B: Moderate, Group C: Marked.

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Preoperative Evaluation (Table 21.1)

Q42. Answer: B — Trifluoroacetic acid

Correct Answer: B. Trifluoroacetic acid — Sevoflurane does not produce Trifluoroacetic acid on metabolism, thereby avoiding any possibility of hepatitis (unlike halothane).

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative – Maintenance

Q43. Answer: B — Lactulose and neomycin

Correct Answer: B. Lactulose and neomycin — Correction of encephalopathy (by lactulose and neomycin) should be done before taking patients for surgery.

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Timing of Surgery

Q44. Answer: B — Tapping of ascites

Correct Answer: B. Tapping of ascites — Tapping of ascites (if significant enough to impair respiratory functions) should be done as part of preoperative optimization.

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Timing of Surgery

Q45. Answer: C — Creatinine

Correct Answer: C. Creatinine — Child-Pugh includes: Bilirubin, Albumin, Prothrombin time, Ascites, Encephalopathy, Nutrition, Surgical risk, and Mortality. Creatinine is part of MELD score, not Child-Pugh.

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Preoperative Evaluation (Table 21.1)

Q46. Answer: B — Drug binding — increasing the free fraction of the drug

Correct Answer: B. Drug binding — increasing the free fraction of the drug — Decreased albumin may increase the unbound fraction of the drug, leading to exaggerated clinical effects of protein-bound drugs.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative – General Anesthesia

Q47. Answer: B — Hepatitis B or C

Correct Answer: B. Hepatitis B or C — Universal precautions must be taken if the patient is positive for hepatitis B or C to protect the operating room staff.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative

Q48. Answer: C — Rheumatoid arthritis

Correct Answer: C. Rheumatoid arthritis — Important to rule out: coagulopathy, renal damage (hepatorenal syndrome), pulmonary dysfunction (hepatopulmonary syndrome), portopulmonary hypertension, encephalopathy, ascites, gastroesophageal varices, cardiac functions, electrolyte and metabolic abnormalities.

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Preoperative Evaluation

Q49. Answer: B — Whatever discussed for hepatic disease is applicable, with additional points to be taken care

Correct Answer: B. Whatever discussed for hepatic disease is applicable, with additional points to be taken care — Whatever discussed for the management of patients with hepatic disease, the same is applicable for patients with biliary obstruction with additional points (bradycardia, mannitol, vecuronium avoidance, opioid caution).

Topic: Anesthesia for Biliary Obstruction | Ref: Ch 21, Page 200, Topic: Anesthesia for Patients with Biliary Obstruction

Q50. Answer: B — Comorbidities including cardiomyopathy, pancreatitis, and nutritional deficiencies

Correct Answer: B. Comorbidities including cardiomyopathy, pancreatitis, and nutritional deficiencies — Always rule out alcoholism and associated comorbidities in hepatic patients.

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Preoperative Evaluation